

SONNEBERG NEUFANG, Germany

WMO: 105580

Lat: 50.383N

Lon: 11.183E

Elev: 2067

StdP: 13.63

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	6.3	11.5	-1.3	5.5	11.7	3.2	7.0	12.9	31.8	23.8	28.0	20.8	12.0	30	0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	79.3	63.4	75.9	62.2	73.0	60.8	65.5	75.9	63.7	72.9	62.0	70.3	7.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.6	88.5	69.0	60.1	83.8	67.0	58.7	79.5	65.7	31.5	76.3	30.1	73.0	28.8	70.9	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.4	20.9	18.2	DB	5.0	85.9	7.4	2.7	-0.3	87.8	-4.6	89.4	-8.8	90.9	-14.2	92.9	(4)
(5)			WB	4.0	69.0	7.4	1.4	-1.3	70.0	-5.6	70.9	-9.8	71.7	-15.1	72.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.2	27.6	27.6	35.4	42.2	51.7	56.4	61.1	61.0	54.0	46.0	35.3	30.7
	DBStd	14.06	8.46	8.33	7.42	7.26	7.36	7.06	6.60	6.38	6.23	6.68	6.63	6.59
	HDD50	3345	694	628	454	251	72	17	2	2	26	160	442	597
	HDD65	7678	1159	1048	917	683	413	269	157	157	331	590	892	1062
	CDD50	1228	0	0	2	17	125	211	347	343	148	35	0	0
	CDD65	86	0	0	0	0	1	12	37	33	3	0	0	0
	CDH74	495	0	0	0	0	4	70	220	193	8	0	0	0
	CDH80	67	0	0	0	0	0	6	32	29	0	0	0	0
Wind	WSAvg	8.2	8.3	9.1	8.9	9.2	9.1	8.0	7.5	7.2	7.7	8.2	8.0	7.2
Precipitation	PrecAvg	36.9	3.5	2.8	2.6	2.1	2.8	3.1	3.9	2.9	3.0	2.8	3.3	3.8
	PrecMax	46.6	8.1	7.4	6.3	4.4	7.0	5.4	5.9	7.7	7.2	9.1	6.0	9.5
	PrecMin	27.0	0.1	0.4	0.5	0.2	0.5	1.0	1.5	1.0	0.6	0.7	0.2	1.2
	PrecStd	5.4	1.7	1.6	1.5	1.0	1.6	1.3	1.2	1.4	1.3	1.7	1.5	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.4	50.8	59.6	66.1	74.3	80.9	84.0	83.7	74.8	66.6	52.4	48.4
		MCWB	43.3	45.2	47.1	52.0	58.3	64.3	65.0	66.2	60.4	57.2	46.6	46.1
	2%	DB	41.6	45.3	53.1	61.5	70.6	75.6	79.7	78.9	71.4	62.0	49.2	44.7
		MCWB	40.7	41.4	45.5	50.3	56.3	61.7	63.7	63.3	59.7	54.8	46.3	42.8
	5%	DB	39.6	41.5	48.6	57.8	67.3	72.2	76.1	75.3	67.4	57.9	46.9	42.2
		MCWB	38.8	38.8	42.9	47.8	54.8	59.8	62.7	62.2	58.2	52.8	44.3	40.9
	10%	DB	37.7	37.7	44.9	53.8	64.0	68.4	72.9	71.7	63.8	54.9	44.0	39.7
		MCWB	36.8	36.0	41.3	45.9	53.5	58.1	61.4	60.9	56.8	50.9	42.2	38.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.1	46.0	50.0	53.6	60.6	66.1	67.9	68.2	63.8	58.7	49.2	46.7
		MCDB	44.8	49.1	55.3	62.7	68.9	77.3	80.0	80.8	69.7	64.3	50.0	47.4
	2%	WB	40.9	42.3	46.2	51.0	57.9	63.4	65.7	65.4	61.2	56.0	47.3	43.6
		MCDB	41.3	44.7	52.2	59.6	66.9	72.6	76.3	75.6	67.6	60.2	48.9	44.2
	5%	WB	39.1	39.0	43.9	48.6	56.1	61.3	63.8	63.4	59.3	53.5	44.8	41.1
		MCDB	39.5	40.9	47.7	55.3	65.2	69.0	73.5	72.4	65.0	57.0	46.2	42.0
	10%	WB	36.9	36.2	41.5	46.6	54.2	59.2	61.9	61.7	57.6	51.5	42.2	39.1
		MCDB	37.5	37.6	44.6	52.8	62.5	66.3	70.9	69.7	62.8	54.4	43.5	39.6
Mean Daily Temperature Range	5% DB	MDBR	6.7	8.2	10.0	13.5	14.3	13.6	14.8	14.4	12.4	10.3	7.2	6.2
		MCDBR	8.8	11.7	16.1	19.4	19.2	18.8	19.5	19.6	17.2	14.2	11.0	8.6
		MCWBR	8.4	8.8	9.9	10.9	10.3	9.7	9.7	9.5	9.1	8.7	8.8	7.5
	5% WB	MCDBR	8.3	10.4	14.0	17.4	17.1	16.9	18.1	17.1	15.3	12.8	10.0	7.6
		MCWBR	8.2	8.6	9.7	10.4	10.1	9.5	10.0	9.3	9.0	8.4	8.7	7.3
Clear-Sky Solar Irradiance	taub	0.286	0.307	0.341	0.375	0.380	0.379	0.391	0.385	0.355	0.333	0.311	0.284	
	taud	2.444	2.406	2.374	2.320	2.330	2.346	2.310	2.351	2.418	2.481	2.489	2.447	
	Ebn at Noon	242	265	273	275	278	279	273	269	266	252	230	225	
	Edh at Noon	19	25	31	37	38	38	39	35	30	23	18	16	
All-Sky Solar Radiation	RadAvg	249	485	824	1298	1541	1686	1599	1381	993	583	276	189	
	RadStd	46	75	108	169	181	142	185	142	140	73	38	23	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.65	N/A	N/A	N/A	N/A

Nomenclature: See separate page